

Second-Price Auction

SPA in Independent Private Value Environments

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This talk

Overview of Main Auction Formats

- English and Dutch auctions
- sealed-bid (first-price and second-price) auctions

Overview of Main Environments

- independent private values (IPV)
- common value (CV)

Second-Price Auction

- truthfulness
- asymmetry: bilateral negotiation vs. SPA

Main Auction Formats and Environments

4 Main Auction Formats for a Single Object

- sealed-bid first price auction (FPA)
- sealed-bid second price auction (SPA)
- ascending-bid (English) auction
- descending-bid (Dutch) auction

Main Statistical Models of Valuations

Independent private values

- statistically independent valuations of the object
- privately observed without noise
- e.g., consumption goods
- M&A context: strategic bidders
 - operating companies, typically in an industry related to the target, e.g., competitors, suppliers, or customers
 - upon acquiring the target, they usually integrate it into their existing operations with the idea of generating value through synergies
 - value through highly bidder-specific synergies
 - primarily interested in running the company, secondarily interested in resale

Main Statistical Models of Valuations

Common value

- an identical (common) valuation of the object
- privately observed with (typically additive and independent) *noise*
- e.g., investment goods or durable consumption goods with a resale option
- M&A context: private equity firms
 - They look for undervalued targets with the potential to improve operations, often with the help of aggressive use of leverage and high-powered managerial incentives.
 - Upon acquiring the target, financial buyers usually keep it as a separate operating company and sell it in a few years to a private buyer or in an IPO.
 - value through leverage and incentives, less from bidder-specific synergies
 - primarily interested in resale, secondarily interested in running the company

A Symmetric IPV Environment

Setup

- 1 indivisible object owned by a *risk-neutral* seller
- N *risk-neutral* buyers without any liquidity/budget constraints
- X_i is buyer i 's stochastic private value of the object
- $X_i \sim F$ are i.i.d. (*independently* and *identically* distributed)

= a *symmetric* IPV environment

- F is continuous, increasing and twice differentiable on an interval $[0, \omega]$

= F admits a continuous density f and has full support: $f(x_i) > 0$ for $x_i \in [0, \omega]$

A Symmetric IPV Environment

Information

- each buyer observes her value X_i
- x_i is a realization of X_i (a specific *type* of player i)
- given independence, $X_i \perp\!\!\!\perp X_j$, the realization x_i is completely uninformative about any other $X_j, j \neq i$

TL;DR: The buyer i learns *everything* about X_i and *nothing* about any other X_j .

SPA in IPV Environment

SPA (Vickrey auction)

- Each player i submits a sealed bid $b_i \geq 0$.
- The highest bidder wins the object and pays the *second* highest bid.
- Take a strategy/bid profile $b = (b_1, \dots, b_N)$.
- 1 winner

$$\Pi_i(x_i, b) = \begin{cases} x_i - \max_{j \neq i} b_j, & \text{if } b_i > \max_{j \neq i} b_j \\ 0, & \text{if } b_i < \max_{j \neq i} b_j. \end{cases}$$

- 2+ winners (a tie)
 - the first (highest) bid is also the second highest bid
 - lottery with equal probability of winning among the highest bidders
 - each of k winners: $\Pi_i(x_i, b) = \frac{1}{k}(x_i - b_i)$

Back to our old example.

Example (Two bidders with *known* asymmetric valuations, discrete bids)

- $v_1 = 10, v_2 = 5$
- SPA has many Nash equilibria, including $b_1 = 0, b_2 = 12$ where price is zero.
 - Here, the stronger player expects an aggressive opponent *with probability 1* and therefore doesn't submit a non-competitive bid.
 - The non-competitive bid becomes the 'market price'.
 - Weaker player expects an attractive price created by a non-competitive bid and to motivate the stronger player to remain non-competitive, she is aggressive.
- Now, what if the weaker player has many types (e.g., $v_2 \in \{4, 5, 6\}$), and at least *some* types are expected to be non-aggressive?
 - Then, the stronger player expects an aggressive opponent *with probability* < 1 . She submits a competitive bid to win at least against non-aggressive types.
 - ! The threat of the weaker player becomes completely ineffective.

How to get a Nash equilibrium in SPA in an IPV environment?

Method 1: Weak dominance

- We find unique weakly undominated strategies.
- For this method, we (i) restrict ourselves by using weak dominance (ii) without the positive density property of the opponents' types (see below).

Method 2: Best responses

- We find a unique best response.
- For this method, we (i) use that each player expects that the highest opponent's value is distributed on $[0, \omega]$ with a positive density (ii) without applying weak dominance.

Method 1: Weak dominance

Preview

- Claim: If only a *single* weakly undominated strategy exists for every player, then the strategy profile with these weakly undominated strategies is a Nash equilibrium.
- Why? For every player, by weak dominance over *any other* strategy, there is no profitable deviation. There, the strategy is a best response strategy. Nash equilibrium is defined as mutual best responses.

Method 1: Weak dominance

Let $z_i = \max_{j \neq i} b_j$ (the realized highest bid among all opponents' bids).

What are best response strategies $B_i(z_i)$ in SPA?

- $z_i < x_i$: any bid $b_i > z_i$, including x_i (positive profits for win), $B_i(z_i) = (z_i, \infty)$
 - $z_i = x_i$: any bid $b_i \geq 0$, including x_i (zero profits for win, tie and loss), $B_i(z_i) = [0, \infty)$
 - $z_i > x_i$: any bid $b_i < z_i$, including x_i (zero profits for loss), $B_i(z_i) = [0, z_i)$
- ! Bidding **own value** is a best response bid (one of many best response bids) irrespective of the opponent's bids: $x_i \in B_i(z_i)$ for any z_i .

Method 1: Weak dominance

- Now take *all* possible opponents' pure strategy profiles.
- Each such profile generates a specific z_i (the highest opponents' bid), where each $z_i \in [0, \omega]$ can be mapped to some strategy profile.
- Now observe:

$$\bigcap_{z_i \in [0, \omega]} B_i(z_i) = \{x_i\}$$

- Given this property, the **truthful** bid $b_i = x_i$ *weakly dominates any other* bid in $[0, \omega]$.
- In SPA, there exists a *unique* equilibrium in weakly undominated (truthful) strategies.

Method 2: Best response against a stochastic opponent's bid

Preview

- We use that in IPV environment, each player expects that the highest opponent's value Z_i is distributed on $[0, \omega]$ with a *positive* density.
- We will see that, if all opponents submit truthful bids, then each player's unique best response is to submit a truthful bid.

Method 2: Best response against a stochastic opponents' bids

- Suppose that the density of the opponents' highest bid is positive for any $z_i \in [0, \omega]$: $h(z_i) > 0$.
- In Nash equilibrium, you have correct belief about $h(z_i)$.
- With a bid b_i , your expected payoff is therefore

$$\int_0^{b_i} (x_i - z_i) h(z_i) dz_i.$$

- The marginal effect of an increase in b_i on the expected payoff is $(x_i - b_i)h(b_i)$.
- Given $h(\cdot) > 0$, the expected payoff is given by the sign of the marginal effect:
 - increasing when $b_i < x_i$
 - decreasing when $b_i > x_i$
 - maximized at $b_i = x_i$.
- Finally, since all players of all types in IPV environment set truthful bids, $b_i(x_i) = x_i$, then we confirm $h(z_i) > 0$ for any $z_i \in [0, \omega]$.

What is the trick?

Think of a low bid $b_i < x_i$. Does it pay off to be more aggressive or not?

- In SPA, increasing the bid implies:
 - + winning profits against marginal opponents
 - 0 stable profits with inframarginal opponents
- In FPA, increasing the bid implies:
 - + winning profits against marginal opponents
 - reducing profits with inframarginal opponents

SPA Fuckups

SPA Fuckup I: Google's Project Bernanke

- The sealed-bid SPA has the disadvantage that the second-price is not immediately observed to the buyer; the buyer/winner directly observes only own bid.
 - ! This price can be manipulated by the *auctioneer*.
- In contrast, in English auction (strategically equivalent to SPA under IPV), the price is immediately publicly observed.
 - ! This is even pronounced when the auctioneer is a (two-side) platform: both the seller as well as buyer can be manipulated.
- Google's Project Bernanke: the buyer paid higher than the second price and the seller received less than the second price.

SPA Fuckup II: New Zealand

New Zealand 1990 spectrum auction was SPA with zero reserve price.

- + efficiency: highest-value users
- + speed
- + low administrative cost

What could go wrong?

- low revenues (only 14% of predicted revenues)
- large differences between the winning bid and prices (politically sensitive), e.g., NZ\$100,000 and NZ\$6; NZ\$7,000,000 and NZ\$5,000

What is the problem?

- low competition due to only a few bidders (unknown number of bidders)
- to be addressed by a reserve price (to 'restore bargaining power' of the seller); we will show that it introduces a tradeoff between ex post efficiency and revenues

SPA in an Asymmetric IPV Environment

SPA in Asymmetric IPV Environment

Suppose that the buyer 1's private value is (on average) relatively larger.

- stronger buyer/bidder $X_1 \sim U[0, 1]$
- weaker buyer/bidder $X_2 \sim U[0, v], v < 1$
- X_1 and X_2 are independent

The stronger buyer is stronger only *ex ante* but not necessarily *ex post*!

SPA in Asymmetric IPV Environment

$$\begin{aligned}\mathbb{E}(\min\{X_1, X_2\}) &= \int_0^v \frac{x_1}{v} \mathbb{E}(x_2 \mid x_2 \leq x_1) + \left(1 - \frac{x_1}{v}\right) x_1 dx_1 + \int_v^1 \frac{v}{2} dx_1 = \\ &= \int_0^v \frac{x_1}{v} \frac{x_1}{2} + \left(1 - \frac{x_1}{v}\right) x_1 dx_1 + \int_v^1 \frac{v}{2} dx_1 = \\ &= \int_0^v x_1 - \frac{x_1^2}{2v} dx_1 + \int_v^1 \frac{v}{2} dx_1 = \\ &= \left[\frac{x_1^2}{2} - \frac{x_1^3}{6v} \right]_0^v + \left[\frac{v}{2} x_1 \right]_v^1 = \frac{v^2}{3} + \frac{v}{2} - \frac{v^2}{2} = \\ &= \frac{v}{2} - \frac{v^2}{6}\end{aligned}$$

- revenues are from zero to $\frac{1}{3}$

Bilateral negotiations or SPA?

- take-it-or-leave-it offer $p \in [0, 1]$ to the (ex ante) stronger buyer
- (expected) revenues are $[1 - F(p)]p = (1 - p)p$
- revenues maximized for the price $p = \frac{1}{2}$, and equal $\frac{1}{4}$
- bilateral negotiation with the stronger buyer *only* is preferred over auction with *both* buyers if the asymmetry is large

$$\frac{1}{4} > \frac{v}{2} - \frac{v^2}{6}$$

$$v < \frac{3 - \sqrt{3}}{2} \doteq 0.63$$

Reference

- Krishna, V. (2010). *Auction Theory*. Elsevier. Chapters 1, 2.1, 2.2.